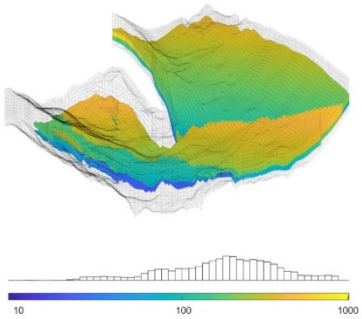
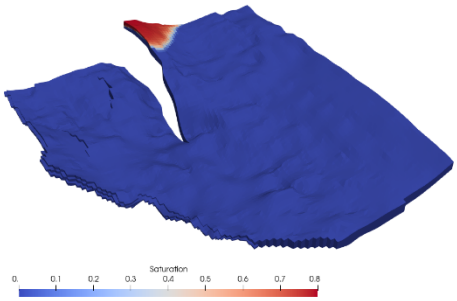
DARSim Example:

Johansen Formation





This example demonstrates how to use corner-point geometry (CPG) in DARSim to simulate complex geological structures. H_2 is injected at the highest point of the NPD5 sector of the Johansen Formation. The figure above shows the sector's permeability distribution (mD) and the resulting gas saturation after five years of H_2 injection at a rate of 1E-5 PV/day.

The Johansen formation is a large saline aquifer in the Northern North Sea, off the South Western coast of Norway. The formation consists of sandstone and has moderate permeability, ranging from 60 to 1660 mD, and an average porosity of around 25%. Further information is available in Eigestad et al. (2009), Hosseinimehr et al. (2021), and van den Dorpel (2025).

To use a CPG grid, set the grid discretization type to CornerPointGrid and list the relevant .grdecl file or files on the following lines:

```
--Grid Discretization ("CartesianGrid" or "CornerPointGrid")
GRID_DISCRETIZATION
      CornerPointGrid
      NPD5_ geometry_and_properties.grdecl
```

The .grdecl files must follow the standard Eclipse grid format and include the SPECGRID, COORD, ZCORN, and ACTNUM keywords. For a homogeneous domain, porosity and permeability may be specified in DARSim's main input file. For a heterogeneous domain, these properties may instead be included in the .grdecl files using the PORO, PERMX, PERMY, and PERMZ keywords.

DARSim processes the .grdecl files and generates a file named CornerPointGridData.mat, which is then used during the simulation. If this file already exists in the working directory, DARSim will use it directly and will not reprocess the .grdecl files. Therefore, after modifying a .grdecl file, delete

CornerPointGridData.mat before rerunning the simulation so that DARSim processes the updated input.